



FOOD FOR THOUGHT

Ryan Fernando



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Why strong bones make stronger champions

While fitness levels among women athletes have soared, the **foundation** of true success still rests on strong, healthy bones.

When India lifted the women's ODI World Cup this year, it was not only a proud moment for the country but also a sign that the new generation of athletes is stronger and better prepared than ever before. Behind such successes are years of science-based nutrition, dedication and a commitment to understanding how to protect and strengthen the skeleton.

I have spent over 20 years working with some of India's finest female athletes, including Harmanpreet Kaur, Smriti Mandhana, P. V. Sindhu, Joshna Chinappa, Poorva Sawant and Aditi Dhumatkar. Across all these years, one subject has always remained at the centre of my work: bone health. It is the foundation that supports performance, endurance and long-term strength, especially for Indian women in sport.

Density dilemma

Indian women, in general, are more likely to have low bone mineral density, due to genetics, reduced sunlight exposure, lower protein intake and cultural dietary habits. Many skip milk because of intolerance or taste, avoid sunlight for fear of tanning, or simply do not eat enough nutritious food during their growing years.

By the time they enter competitive sport, their bones already carry a disadvantage. Once training intensifies and workloads rise, the risk of stress fractures and joint pain increases. Several promising athletes lose precious time because their bones were not prepared for the demands of elite sport. The good news is that with awareness and the right diet, this can be corrected.

Let's start with the basics — the composition of bone.

Bones are made up of about 60-70 per cent minerals (mainly calcium and phosphorus), 20-25 per cent collagen and around 10-15 per cent water. Collagen forms the framework that gives bones their flexibility and shock absorption, while minerals provide strength and rigidity. I have even seen the difference first-hand. One of my athletes once suffered a bone injury, and after we increased collagen intake, the recovery was noticeably faster than expected. It reinforced my belief in collagen's crucial role in bone repair and resilience.

The sunshine gap

Nearly 80 per cent of Indian athletes I have tested show low levels of Vitamin D. This is surprising in a country that has sunlight all year. The reason is simple: many players train indoors, travel in buses and cover their skin with clothing or sunscreen. Vitamin D is vital for calcium absorption and overall bone function.

I recommend natural sunlight exposure in the morning for at least 20 minutes without sunscreen. I also encourage including fortified foods in the diet. In some cases, I guide